# PH3244
# Experiment - 5
# The Cornus method of determining the elastic constants of a transparent material

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## Synopsis

In this experiment, we use Cornu's method to determine the Young's moduls and Possion's ratio of a given acryllic slab.

CONTENTS

**I. Theory and Procedure** 3
A. Theory 3
B. Procedure 4

**II. Observation** 4
1. Slab parameters and other values 4
2. Fringe Readings 5

**III. Calculations, Analysis and Characteristics** 5
A. Sources of Error 5
1. Systematic Errors 5
2. Random Errors 5
B. Error Propogation 5
1. Calculating error from least count 5
2. Error propogation for $R_0, Y$ and $\zeta$ 6
3. Error propogation for $R_1$ 6
4. Error propogation for $R_2$ 7
C. Plots and Calculation 7

**IV. Result and Conclusion** 9

**References** 10

**A. Theoretical values and Uncertainty** 10

**B. Imporovments in procedure** 10

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## I. THEORY AND PROCEDURE

### A. Theory[1]

The radius of curvature of plano convex lens $R_0$ is given by:

$$R_0 = \frac{d_n^2}{4n\lambda} \tag{1}$$

where the $d_n^2$ is the diameter of $n^{\text{th}}$ fringe, $\lambda$ is the wavelength used. The $d_n^{'2}$ and $d_n^{''2}$ along with $d_n^2$ can be seen properly in FIG - 1.

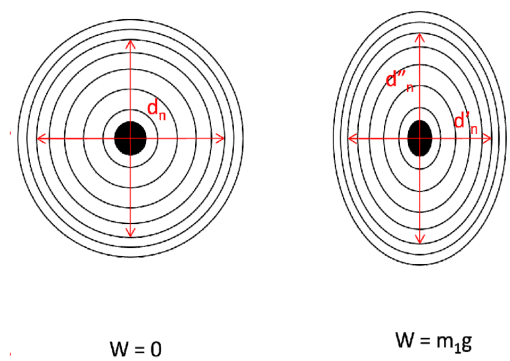


FIG. 1: Newton rings with and without bending shown, respectively, in the left and right panel.

The Young's moduls ($Y$) and the Possion's ratio ($\zeta$) are calculated by:

$$Y = \frac{12m_1gLR_1}{wh^3} \tag{2}$$

$$\zeta = \frac{R_1}{R_2} \tag{3}$$

Here $m_1$ is the mass on each side of the bar, $g$ is the gravitational acceleration, $L$ is the length of the beam from the pivot to a end, $w$ and $h$ are width and height of the road respectively and can be seen from FIG - 2.

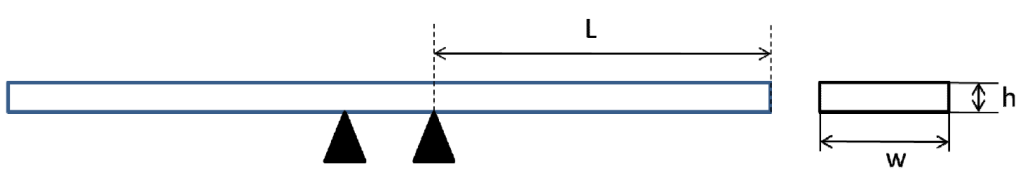


FIG. 2: Side views (the unloaded condition) of the beam whose elastic constants have to be determined. The beam is supported on two triangular pegs (black triangular shape objects in the figure)

[1] Please refer to [Cat (2026)] for a full examination of theory, only the important results will be discussed here.

Here $R_1$ and $R_2$ are obtained from the equations:

$$\frac{1}{R_1} = 4n\lambda\left(\frac{1}{d_n'^2} - \frac{1}{d_n^2}\right) \tag{4}$$

$$\frac{1}{R_2} = 4n\lambda\left(\frac{1}{d_n''^2} - \frac{1}{d_n^2}\right) \tag{5}$$

and are two differnet radius of curvature which can be seen from FIG - 3.

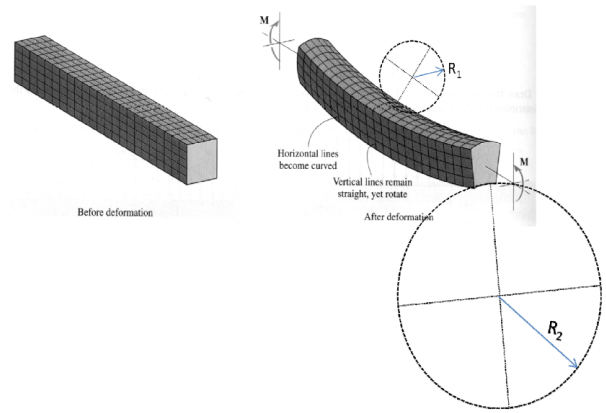


FIG. 3: Longitudinal bending ($R_1$) of a bean resulting in small lateral bending ($R_2$)

### B. Procedure [2]

From Equation - 1, 4 and 5, we can determine the values of $R_0$, $R_1$ and $R_2$ from the slope obtained by plotting the graph of $n$ vs different values form those radius of curvatures, form Equation - 2 and 3 we can determine what Youngs modulus and Possion's ratio of the given material is.

## II. OBSERVATION

### 1. Slab parameters and other values[3]

$$\begin{aligned} w &= 41\text{mm} \\ l &= 125\text{mm} \\ h &= 3.7\text{mm} \\ g &= 9.8 * 10^3 \text{mm/s}^2 \\ \lambda &= 589 * 10^{-6}\text{mm} \end{aligned}$$

[2] Refer to the manual [Cat (2026)] for a look at procedure.
[3] The least count is to be interpreted from the significant figures written

### 2. Fringe Readings[4]

| Left | | Right | |
|---|---|---|---|
| M.S. | C.S. | M.S. | C.S. |
| 12.5 | 19 | 13.5 | 41 |
| 12.5 | 9 | 13.5 | 49 |
| 12.5 | 0 | 14.0 | 19 |
| 13.0 | 43 | 14.0 | 26 |
| 13.0 | 35 | 14.0 | 34 |
| 13.0 | 29 | 14.0 | 44 |
| 13.0 | 24 | 14.0 | 44 |

TABLE I: Readings for $m_1 = 0$g.

| Left | | Right | | Top | | Bottom | |
|---|---|---|---|---|---|---|---|
| M.S. | C.S. | M.S. | C.S. | M.S. | C.S. | M.S. | C.S. |
| 13.5 | 18 | 15.0 | 31 | 11.0 | 6 | 11.5 | 14 |
| 13.5 | 7 | 14.0 | 40 | 10.5 | 37 | 11.5 | 25 |
| 13.5 | 0 | 14.0 | 49 | 10.5 | 29 | 11.5 | 33 |
| 13.0 | 43 | 14.5 | 6 | 10.5 | 21 | 11.5 | 39 |
| 13.0 | 38 | 14.5 | 12 | 10.5 | 15 | 11.5 | 46 |
| 13.0 | 32 | 14.5 | 17 | 10.5 | 9 | 12.0 | 4 |

TABLE II: Readings for $m_1 = 98$g.[a]

[a] On one side its 97g and the other 99g, hence taking avgerage we get $m_1 = 98$g.

| Left | | Right | | Top | | Bottom | |
|---|---|---|---|---|---|---|---|
| M.S. | C.S. | M.S. | C.S. | M.S. | C.S. | M.S. | C.S. |
| 13.5 | 13 | 14.0 | 21 | 10.0 | 40 | 11.0 | 6 |
| 13.5 | 3 | 14.0 | 31 | 10.0 | 28 | 11.0 | 18 |
| 13.0 | 44 | 14.0 | 39 | 10.0 | 20 | 11.0 | 25 |
| 13.0 | 37 | 14.0 | 46 | 10.0 | 13 | 11.0 | 33 |
| 13.0 | 32 | 14.5 | 2 | 10.0 | 7 | 11.0 | 40 |
| 13.0 | 27 | 14.5 | 7 | 10.0 | 1 | 11.0 | 46 |

TABLE III: Readings for $m_1 = 197$g.

| Left | | Right | | Top | | Bottom | |
|---|---|---|---|---|---|---|---|
| M.S. | C.S. | M.S. | C.S. | M.S. | C.S. | M.S. | C.S. |
| 12.5 | 43 | 13.0 | 46 | 10.0 | 40 | 11.0 | 6 |
| 12.5 | 32 | 13.5 | 7 | 10.0 | 28 | 11.0 | 17 |
| 12.5 | 23 | 13.5 | 16 | 10.0 | 19 | 11.0 | 25 |
| 12.5 | 16 | 13.5 | 22 | 10.0 | 12 | 11.0 | 34 |
| 12.5 | 11 | 13.5 | 28 | 10.0 | 6 | 11.0 | 40 |
| 12.5 | 6 | 13.5 | 33 | 9.5 | 44 | 11.0 | 46 |

TABLE IV: Readings for $m_1 = 296$g.

Data taken on 16 Feb 2026.

## III. CALCULATIONS, ANALYSIS AND CHARACTERISTICS

### A. Sources of Error

#### 1. Systematic Errors

1. The weight of the slab on itself.
2. Measureing chord instead of Diameter.
3. Unequal Weight on both sides.
4. Index of fringe being differnet than what is measured.

#### 2. Random Errors

1. Human errros in measurement.
2. Uncertanity in instruments.
3. Shaking of table resulting in moving fringes.

### B. Error Propogation

#### 1. Calculating error from least count

To calculate the standard deviation from values we have measured with some least count, and the values we have applied digitally, we use the formula [Tewari (2016)]:

[4] M.S. Stands for Main Scale Readings and C.S. Stands for Circulat Scale Readings

$$\sigma = \frac{\text{Least Count}}{\sqrt{12}} \tag{6}$$

and to calculate the standard deviation for values we have applied on analog instrument with some least count, we use the formula [Tewari (2016)]:

$$\sigma = \frac{\text{Least Count}}{\sqrt{24}} \tag{7}$$

## 2. Error propogation for $R_0, Y$ and $\zeta$

Now to calculate $R_0$, from general error propogation formulas we see [Taylor (1997)] that:

$$\boxed{\sigma_{d_n^2} = \sqrt{2} d_n \sigma_{d_n} \implies \sigma_{d_n^2} = \frac{\text{Least Count}}{\sqrt{6}}} \tag{8}$$

Now for $Y$ and $\zeta$ from Equation 2, 3 we see [Taylor (1997)] that:

$$\sigma_Y = Y\sqrt{\left(\frac{\sigma_{m_1}}{m_1}\right)^2 + \left(\frac{\sigma_g}{g}\right)^2 + \left(\frac{\sigma_L}{L}\right)^2 + \left(\frac{\sigma_{R_1}}{R_1}\right)^2 + \left(\frac{\sigma_w}{w}\right)^2 + \left(3\frac{\sigma_h}{h}\right)^2} \tag{9}$$

$$\sigma_\zeta = \zeta\sqrt{\left(\frac{\sigma_{R_1}}{R_1}\right)^2 + \left(\frac{\sigma_{R_2}}{R_2}\right)^2} \tag{10}$$

## 3. Error propogation for $R_1$

To calculate $R_1$ and $R_2$ we see that we need to plot $n$ vs $\frac{x^2y^2}{|x^2-y^2|}$ (From Equation 4, 5). Let the uncertainty in $d_n$, $d'_n$, and $d''_n$ be : $\sigma_{d_n}$, $\sigma_{d'_n}$, and $\sigma_{d''_n}$ respectively. For a function $f(x, y)$ with independent variables, the propagated error is

$$\sigma_f^2 = \left(\frac{\partial f}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 \sigma_y^2$$

Let

$$x = d_n, \qquad y = d'_n$$

$$f = \frac{x^2y^2}{x^2 - y^2}$$

Using the quotient rule,

$$\frac{\partial f}{\partial x} = \frac{2xy^2(x^2 - y^2) - x^2y^2(2x)}{(x^2 - y^2)^2}$$

Simplifying,

$$\frac{\partial f}{\partial x} = \frac{-2xy^4}{(x^2-y^2)^2}$$

Similarly,

$$\frac{\partial f}{\partial y} = \frac{2yx^4}{(x^2-y^2)^2}$$

Substituting into the propagation formula,

$$\sigma_f^2 = \left(\frac{-2xy^4}{(x^2-y^2)^2}\right)^2 \sigma_x^2 + \left(\frac{2yx^4}{(x^2-y^2)^2}\right)^2 \sigma_y^2$$

$$\sigma_f^2 = \frac{4x^2y^8}{(x^2-y^2)^4}\sigma_x^2 + \frac{4y^2x^8}{(x^2-y^2)^4}\sigma_y^2$$

Thus

$$\sigma_f = \frac{2x^2y^2}{(x^2-y^2)^2}\sqrt{y^4\sigma_x^2 + x^4\sigma_y^2}$$

Replacing $x$ and $y$,

$$\boxed{\sigma_f = \frac{2d_n^2 d_n'^2}{(d_n^2 - d_n'^2)^2}\sqrt{d_n'^4\sigma_{d_n}^2 + d_n^4\sigma_{d_n'}^2}} \tag{11}$$

#### 4. Error propogation for $R_2$

Doing the exact same procedure as before and for $f = \dfrac{d_n^2 d_n''^2}{d_n''^2 - d_n^2}$ we get:

$$\boxed{\sigma_f = \frac{2d_n^2 d_n''^2}{(d_n''^2 - d_n^2)^2}\sqrt{d_n''^4\sigma_{d_n}^2 + d_n^4\sigma_{d_n''}^2}} \tag{12}$$

### C. Plots and Calculation[5]

First step is to see weather the data makes iniuive sence or not. for this we plot Table 1 - 4 (we will not include uncertainty in this plot as we only want to see the characteristic) as follows:

[5] Refer to [Cat (2026)] for the code used to plot and determine the slope.

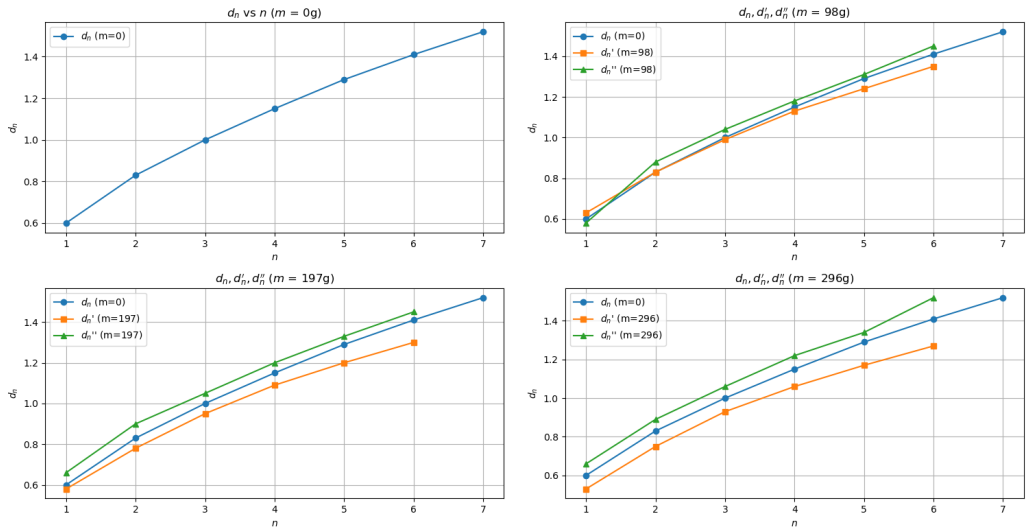


FIG. 4: Plot of Talble 1 - 4 with $d_n$ to compare

From this we see that other than Table - 2 (Top right figure) every other plots seems fine. In the Table two, thhr Hence in our later calcualtiosn we will not consider data from Table - 2 (The general expected order should be $d_n'' > d_n > d_n' r$ but we dont see it for the inital values).
Now using Table - 1 and Equation - 8 for uncertainty, we plot $n$ vs $d_n^2$ as:

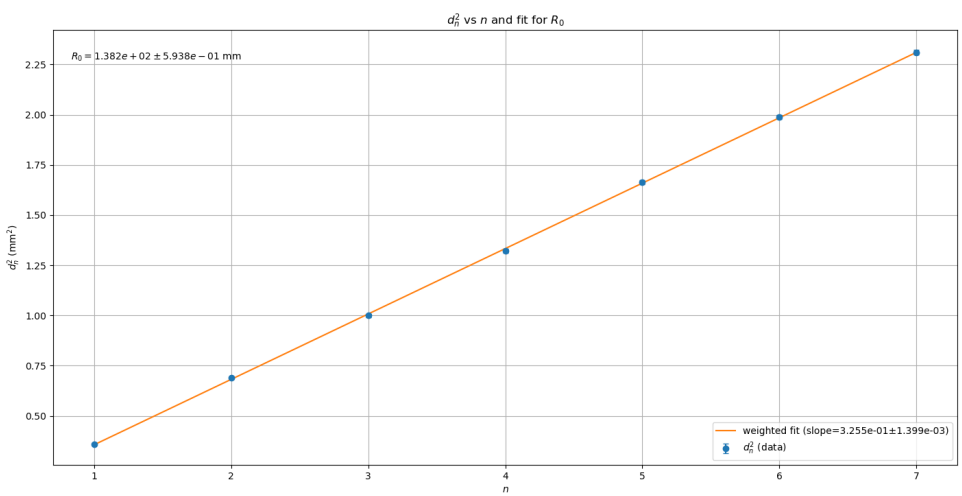


FIG. 5: Linear best fit with error to calculate $R_0$

Now to calcualte the $R_1$ and $R_2$ we use the Equation 4, 5 to calculate it and Equation 10, 11 for its error. Doing it we get:

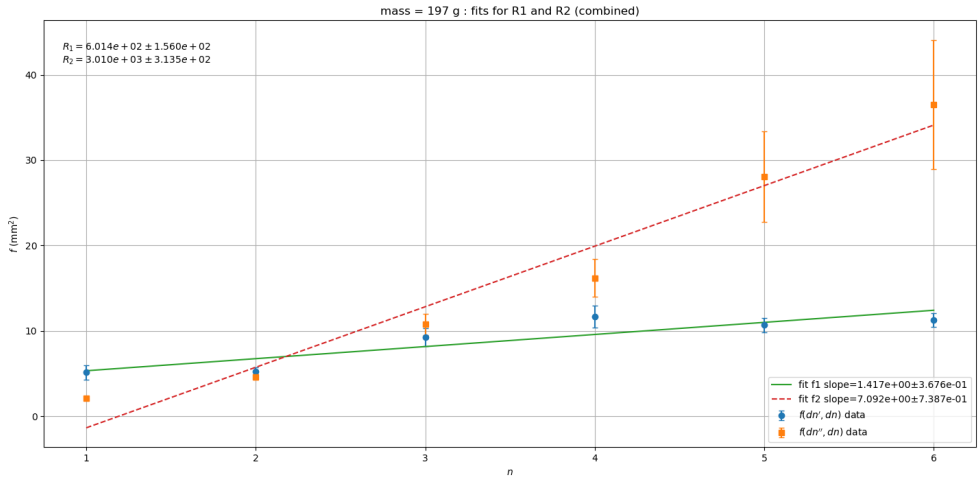


FIG. 6: Plot for $m_1 = 197$g to calculate $R_1$ and $R_2$

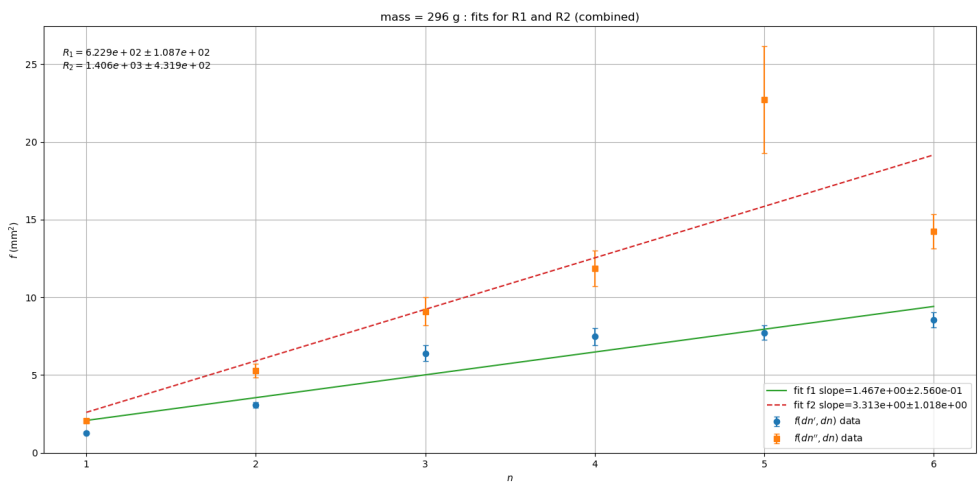


FIG. 7: Plot for $m_1 = 296$g to calculate $R_1$ and $R_2$

From FIG. 6 and 7, we see that the calculated value of $Y$ and $\zeta$ for each mass. Doing a weighted sum we get, $Y = 1.05145 * 10^9 \pm 3.23540 * 10^8$ g s$^{-1}$ mm$^{-2}$.

## IV. RESULT AND CONCLUSION

Hence after converting to SI, the Young's modulus will have unit of Pascals and Possion's ratio will be unitless. The final values we get are:

$$\boxed{\begin{aligned} \text{Young's modulus } (Y) &= 1.0 \pm 0.3 \text{ GPa} \\ \text{Poisson ratio } (\mu) &= 0.325 \pm 0.17 \end{aligned}}$$

The error in these values along with the theoretical value will be discussed in Appendix A.

## REFERENCES

L. M. Cat, Ph3244 3rd year sem 6 physics lab repo: https://github.com/laughingcatmeme/ph3244 (2026).
G. M. Tewari, Measurement uncertainty, International Workshop and Training Program on Good Food Laboratory Practices (2016).
J. R. Taylor, *An Introduction to Error Analysis* (1997).

### Appendix A: Theoretical values and Uncertainty

Perspex (PMMA) acrylic typically exhibits a Young's modulus of 2.4–3.3 GPa (2400–3300 MPa) and a Poisson's ratio of 0.35–0.4. The cause of the deviation for Young's modulus is attributed to very small sample of masses taken. The Possion's ratio fits well within the theoretical value although it has very high uncertainty again because of low sample size.

### Appendix B: Imporovments in procedure

The uncertainty can be decreased by taking more weight measurements. Furthermore to reduce human error we can take multiple measurements for each mass. We can also reduce some systematic errors, such as the weight of the slab, by measuring it and taking it into account for calculations. To have correct fringe index, we can take a picture to determine the peaks, this will be more accurate than eye and can also be double checked later.